Triangle Congruence: SSS and SAS Postulates

Geometry Worksheet · Grade 8–10

Name: _____ Date: _____

Learning Objectives

- Identify and apply the Side-Side-Side (SSS) Postulate to prove triangle congruence
- Identify and apply the Side-Angle-Side (SAS) Postulate to prove triangle congruence
- Write two-column proofs using SSS and SAS postulates, including the Reflexive Property of Congruence

Problems

1. State the SSS Postulate in your own words: if _____ sides of one triangle are congruent to _____ sides of another triangle, then the two triangles are congruent.

2. Given that triangle ABC has sides AB = 5, BC = 7, and AC = 9, and triangle DEF has sides DE = 5, EF = 7, and DF = 9, which postulate proves the triangles are congruent?

$$AB \cong DE,\ BC \cong EF,\ AC \cong DF$$

3. In a diagram, triangles PQR and STU share the property that PQ ≅ ST and QR ≅ TU, but PR and SU are not marked congruent. Can you conclude the triangles are congruent by SSS? Explain.

4. Two triangles share a common side CB. Write the statement and reason that justifies CB ≅ CB in a two-column proof.

$$CB \cong CB$$

5. In quadrilateral ABCD with diagonal CB, you are given AB ≅ CD and AC ≅ BD. Name the third pair of congruent sides needed to prove $\Delta ABC \cong \Delta DCB$ by SSS.

$$CB \cong CB$$

6. Complete the two-column proof: Given AB ≅ CD and AC ≅ BD in quadrilateral ABCD with diagonal CB,

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$\triangle ABC \cong \triangle DCB$

7. State the SAS Postulate: if two sides and the _____ angle between them in one triangle are congruent to two sides and the _____ angle of another triangle, then the triangles are congruent.

8. In triangles XYZ and MNP, you are given XY ≅ MN, ∠Y ≅ ∠N, and YZ ≅ NP. Which postulate proves ΔXYZ ≅ ΔMNP, and why?

$XY \cong MN,\ \angle Y \cong \angle N,\ YZ \cong NP$

9. Determine whether ΔABC ≅ ΔDEF given AB = 8, BC = 10, ∠B = 45°, DE = 8, EF = 10, and ∠E = 45°. State the postulate used.

$AB = DE = 8,\ BC = EF = 10,\ \angle B = \angle E = 45^\circ$

10. In the diagram, M is the midpoint of both diagonals AC and BD of quadrilateral ABCD (i.e., AM ≅ MC and BM ≅ MD), and ∠AMB ≅ ∠CMD. Write a complete two-column proof to show ΔAMB ≅ ΔCMD, stating which postulate applies.

$\triangle AMB \cong \triangle CMD$

Triangle Congruence: SSS and SAS Postulates — Answer Key

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Answer Key

1. Answer: three; three
- Recall the SSS (Side-Side-Side) Postulate definition.
- If all three sides of one triangle are congruent to all three corresponding sides of another triangle, then the triangles are congruent.

2. Answer: SSS Postulate
- Check all three pairs of corresponding sides: AB = DE = 5, BC = EF = 7, AC = DF = 9.
- Since all three pairs of corresponding sides are congruent, apply the SSS Postulate to conclude $\Delta ABC \cong \Delta DEF$.

3. Answer: No; all three pairs of sides must be congruent for SSS.
- SSS requires all three pairs of corresponding sides to be congruent.
- Only two pairs are marked congruent, so SSS cannot be applied.

4. Answer: Statement: CB ≅ CB; Reason: Reflexive Property of Congruence
- Recognize that CB is a shared (common) side between the two triangles.
- Any segment is congruent to itself by the Reflexive Property of Congruence.

5. Answer: CB ≅ CB (Reflexive Property)
- The given information provides AB ≅ CD and AC ≅ BD.
- The diagonal CB is shared by both triangles, so CB ≅ CB by the Reflexive Property, giving the third pair.

6. Answer: 1) AB ≅ CD (Given); 2) AC ≅ BD (Given); 3) CB ≅ CB (Reflexive); Conclusion: $\Delta ABC \cong \Delta DCB$ by SSS
- Statement 1: AB ≅ CD — Reason: Given.
- Statement 2: AC ≅ BD — Reason: Given.
- Statement 3: CB ≅ CB — Reason: Reflexive Property of Congruence.
- Conclusion: $\Delta ABC \cong \Delta DCB$ — Reason: SSS Postulate.

7. Answer: included; included
- Recall the SAS (Side-Angle-Side) Postulate definition.
- The angle must be the included angle — the angle formed between the two congruent sides.

8. Answer: SAS Postulate; ∠Y is the included angle between XY and YZ.
- Identify the two pairs of congruent sides: XY ≅ MN and YZ ≅ NP.
- The angle ∠Y (≅ ∠N) is between sides XY and YZ, making it the included angle.
- Apply the SAS Postulate to conclude $\Delta XYZ \cong \Delta MNP$.

9. Answer: Yes, ΔABC ≅ ΔDEF by the SAS Postulate.

- Identify the two pairs of congruent sides: AB = DE = 8 and BC = EF = 10.
- The included angles are equal: ∠B = ∠E = 45°.
- Since two sides and the included angle are congruent, apply the SAS Postulate.

10. Answer: ΔAMB ≅ ΔCMD by the SAS Postulate.

- Statement 1: AM ≅ MC — Reason: M is the midpoint of AC (Given).
- Statement 2: BM ≅ MD — Reason: M is the midpoint of BD (Given).
- Statement 3: ∠AMB ≅ ∠CMD — Reason: Given (or Vertical Angles Theorem if applicable).
- Conclusion: ΔAMB ≅ ΔCMD — Reason: SAS Postulate (two sides and the included angle are congruent).

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